Lecture class 06.05.25

NO CLASS TOMORROW

* ~~Hard. 11th 12-13:30 11.06~~

* Wed 13th 15:45-17:15 11.33

EX class (as usual) + lecture class (to compensate)

$$(NL)\begin{cases}\dot{x} = f(x) \\ \mathbb{R}^N \ni \bar{x} : f(\bar{x}) = 0\end{cases} \quad , f: \mathbb{R}^N \longmapsto \mathbb{R}^N, N \geq 1 \quad f \in C^1$$

Thm (Lyapunov thm)

$\exists B(\bar{x}), \exists W: B(\bar{x}) \longmapsto \mathbb{R}$ s.t.

1) $W(\bar{x}) = 0$, $W(x) > 0$ $\forall x \in B(\bar{x}) \setminus \{\bar{x}\}$

2) $\dot{W}(x) \leq 0$, $\forall x \in B(\bar{x})$

$\Rightarrow \bar{x}$ is stable for (NL)

If

3) $\dot{W}(x) < 0 \quad \forall x \in B(\bar{x}) \setminus \{\bar{x}\}$

$\Rightarrow \bar{x}$ is asymptotically stable for (NL)

Corollary

$\exists B(\bar{x}), \exists W : B(\bar{x}) \longmapsto \mathbb{R}$ s.t.

1) $W(\bar{x}) = 0, \; W(x) > 0, \; \forall x \in B(\bar{x}) \setminus \{\bar{x}\}$

2') $\dot{W}(x) \geq 0, \forall x \in B(\bar{x})$

3') $\dot{W}(x) > 0, \forall x \in B(\bar{x}) \setminus \{\bar{x}\}$

$\Rightarrow \bar{x}$ is unstable for (NL).

APPLICATION: Lotka-Volterra predator-prey model.

$N = 2$

x : pop. of type 1 $\quad x = x(t)$

y : " " 2 $\quad y = y(t)$

2-dim ODEs system

$$\begin{cases} \dot{x} = x\left(r_x - r_{11}x + r_{12}\theta_x y\right) \\ \dot{y} = y\left(r_y + r_{21}\theta_y x - r_{22}y\right) \end{cases}$$

$$\begin{cases} x(0) = x_0 \geq 0 \\ y(0) = y_0 \geq 0 \end{cases} \Rightarrow \begin{matrix} x(t) \geq 0 \\ y(t) \geq 0 \end{matrix} \quad \forall t \geq 0$$

$r_x, r_y \geq 0$ linear growth rate

$r_{ii} \geq 0$, $i = 1,2$ intra-specific competition rate

$r_{ij} \geq 0$ $i \neq j$ $i,j = 1,2$ inter-specific interaction rate

$\theta_x, \theta_y \in \{-1, 0, 1\}$

1) $\theta_x \cdot \theta_y = 1$

1.1) $\theta_x = \theta_y = 1$ $\rightsquigarrow + r_{12}xy \geq 0$ $+ r_{21}xy \geq 0$

$\Rightarrow$ mutualism

1.2) $\theta_x = \theta_y = -1$ $\Rightarrow$ competition

2) $\theta_x \cdot \theta_y = -1 \Rightarrow$ predator-prey interaction or parasitism interaction.

3) $\theta_x \cdot \theta_y = 0 \Rightarrow$ depending on the cases

Ex $\theta_x = 0$, $\theta_y \neq 0$

$\theta_x = \theta_y = 0 \Rightarrow$ neutral interact.

Predator-prey Lotka-Volterra model

$$(NL_{LV}) \begin{cases} \dot{x} = x(a - by) \\ \dot{y} = y(cx - d) \end{cases} \quad a, b, c, d > 0$$

x: predators

y: preys.

$$(x(0), y(0)) = (x_0, y_0)$$

1) Steady states

$(x_1, y_1) = (0, 0) \rightsquigarrow$ total extinction

$(x_2, y_2) = \left(\frac{d}{c}, \frac{a}{b}\right) \rightsquigarrow$ coexistence

By linearisation

$$J(x,y) = \begin{pmatrix} a-by & -bx \\ cy & cx-d \end{pmatrix}$$

$$J(x_1,y_1) = J(0,0) = \begin{pmatrix} a & 0 \\ 0 & -d \end{pmatrix}$$

$\lambda_1 = a > 0$
$\lambda_2 = -d < 0$
$\Rightarrow$ $(0,0)$ is unstable saddle point for (NL_{LV}) (by Lyapunov criterion)

$$J(x_2,y_2) = J\left(\frac{d}{c}, \frac{a}{b}\right) = \begin{pmatrix} 0 & -bd/c \\ ac/b & 0 \end{pmatrix}$$

$\operatorname{tr} J(x_2,y_2) = 0 \iff \lambda_1 + \lambda_2 = 0 \quad \lambda_1 = -\lambda_2$

$\lambda_1, \lambda_2 \in \rho(J(x_2,y_2))$

$\det J(x_2,y_2) = ad > 0$

$$\det(Jf(x_2,y_2) - \lambda I_2) = \lambda^2 + ad \overset{!}{=} 0$$

$\lambda_{1,2} = \pm i\sqrt{ad}$

$\Longrightarrow (x_2, y_2)$ is a center ONLY for the linearised system

STRATEGY: use the Lyapunov thm.

Lemma

$\bar{z} := \left(\frac{d}{c}, \frac{a}{b}\right)$ is a stable steady state for (NL_{LV})

Proof

$W(x,y) := F(x) + G(y) + W_0$, $W_0 \in \mathbb{R}$

* $\frac{d}{dt} W(x,y) = \dot{W}(x,y) = \partial_1 W \dot{x} + \partial_2 W \dot{y} \Leftarrow$

with (x,y) sol of (NL_{LV})

$\Leftarrow F' \dot{x} + G' \dot{y} = F' x(a - by) + G' y(cx - d) \overset{!}{=} 0$

$$\frac{F'(x) \cdot x}{cx - d} = -\frac{G'(y)\, y}{a - by}$$

$(x,y) \neq \{(0,0), (\bar{x},\bar{y})\}$

$$\frac{F'(x)\cdot x}{cx-d} = 1$$

$$\frac{-G'(y)\cdot y}{a-by} = 1$$

$$F'(x) = c - \frac{d}{x} \Rightarrow F(x) = cx - d\ln(x)$$

$$-G'(y) = \frac{a-by}{y} = \frac{a}{y} - b$$

$$G'(y) = b - \frac{a}{y} \Rightarrow G(y) = by - a\ln(y)$$

$$W(x,y) = cx + by - d\ln(x) - a\ln(y) + W_0$$

$$* \quad W(\bar{x},\bar{y}) \overset{!}{=} 0 \qquad (\bar{x},\bar{y}) = \left(\frac{d}{c}, \frac{a}{b}\right)$$

$$0 = d + a - d\ln\left(\frac{d}{c}\right) - a\ln\left(\frac{a}{b}\right) + W_0$$

$$\Rightarrow W_0 = -(d+a) + d\ln\left(\frac{d}{c}\right) + a\ln\left(\frac{a}{b}\right)$$

$\Rightarrow$ Finally

$$W(x,y) = cx + by - d\ln(x) - a\ln(y) + \\ - (d+a) + d\ln\left(\frac{d}{c}\right) + a\ln\left(\frac{a}{b}\right)$$

* $W(x,y) > 0 \quad \forall \ (x,y) \in B\left((\bar{x},\bar{y})\right) \setminus \{(\bar{x},\bar{y})\}$

we compute — to prove $\rightarrow$ $W(x+\varepsilon_x, y+\varepsilon_y) > 0$, $\varepsilon_x, \varepsilon_y \gtrless 0$

$\varepsilon_x, \varepsilon_y \in \mathbb{R}$

$$W(\bar{x}+\varepsilon_x, \bar{y}+\varepsilon_y) = c\bar{x} + c\varepsilon_x + b\bar{y} + b\varepsilon_y \\ - d\ln(\bar{x}+\varepsilon_x) - a\ln(\bar{y}+\varepsilon_y) + \\ - (d+a) + d\ln(d/c) + a\ln(a/b)$$

$=$ ↑

$\pm\, d\ln(d/c)$

$\pm\, a\ln(a/b)$

$$-(d+a) + d\ln(d/c) + a\ln(a/b) = -F(d/c) - G(a/b)$$

$$F(x) = cx - d\ln(x) \Rightarrow -F(d/c) = -d + d\ln\left(\frac{d}{c}\right)$$

$$G(y) = by - a\ln(y) \Rightarrow -G(a/b) = -a + a\ln\left(\frac{a}{b}\right)$$

$$= c\bar{x} + c\varepsilon_x + b\bar{y} + b\varepsilon_y + d\ln\left(\frac{d}{c}\right) - d\ln\left(\frac{d}{c}\right)$$
$$- d\ln(\bar{x} + \varepsilon_x) - a\ln(\bar{y} + \varepsilon_y) + a\ln\left(\frac{a}{b}\right)$$
$$- a\ln\left(\frac{a}{b}\right) - F(d/c) - G(a/b)$$

$W(\bar{x}, \bar{y}) = 0$

$$= \boxed{d + a - F(d/c) - G(a/b)} \qquad (\bar{x}, \bar{y}) = \left(\frac{d}{c}, \frac{a}{b}\right)$$

$$c\varepsilon_x + b\varepsilon_y + d\ln\left(\frac{d}{c}\right) \boxed{- d\ln\left(\frac{d}{c}\right)}$$
$$- d\ln\left(\frac{d}{c} + \varepsilon_x\right) - a\ln\left(\frac{a}{b} + \varepsilon_y\right)$$
$$\boxed{+ a\ln\left(\frac{a}{b}\right) - a\ln\left(\frac{a}{b}\right)}$$

$$= c\varepsilon_x + b\varepsilon_y + d\ln\left(\frac{d}{c} \cdot \frac{1}{\frac{d}{c} + \varepsilon_x}\right)$$
$$+ a\ln\left(\frac{a}{b} \cdot \frac{1}{\frac{a}{b} + \varepsilon_y}\right)$$

$$= c\varepsilon_x + b\varepsilon_y + d\ln\left(\frac{d}{d+c\varepsilon_x}\right) +$$

$$+ a\ln\left(\frac{a}{a+b\varepsilon_y}\right) =$$

$\varepsilon_x, \varepsilon_y \gtrless 0$ are s.t. $d+c\varepsilon_x > 0$

$a+b\varepsilon_y > 0$

($\varepsilon_x, \varepsilon_y$ are small enough)

$$= c\varepsilon_x + b\varepsilon_y + d\ln\left(1 - \frac{c\varepsilon_x}{d+c\varepsilon_x}\right)$$

$$+ a\ln\left(1 - \frac{b\varepsilon_y}{a+b\varepsilon_y}\right) \approx$$

Recall:

$|x| << 1$: $\ln(1+x) \approx x$

$$\approx c\varepsilon_x + b\varepsilon_y + d\left(-\frac{c\varepsilon_x}{d+c\varepsilon_x}\right) +$$

$$+ a\left(-\frac{b\varepsilon_y}{a+b\varepsilon_y}\right)$$

$$= c\varepsilon_x + b\varepsilon_y - \frac{d\varepsilon_x}{d/c + \varepsilon_x} - \frac{a\varepsilon_y}{a/b + \varepsilon_y}$$

$$= \frac{d\varepsilon_x + c\varepsilon_x^2 - d\varepsilon_x}{d/c + \varepsilon_x} + \frac{a\varepsilon_y + b\varepsilon_y^2 - a\varepsilon_y}{a/b + \varepsilon_y}$$

> 0 ($\varepsilon_x, \varepsilon_y$ small enough and s.t. $d + \varepsilon_x \cdot c > 0$, $a + b\varepsilon_y > 0$)

Summary:

$$W(x,y) := cx + by - d\ln(x) - a\ln(y) + \\ - F(d/c) - G(a/b)$$

s.t. 1) $W(\bar{x},\bar{y}) = 0$

$W(x,y) > 0, \forall (x,y) \in B \setminus \{(\bar{x},\bar{y})\}$

2) $\dot{W}(x,y) = 0$ with (x,y) sol of (NL_{LV})

$\Rightarrow W(x(t),y(t)) = W(x_0,y_0) \quad \forall t \geq 0$

$\Rightarrow$ by Lyap. thm

$\left(\frac{d}{c}, \frac{a}{b}\right)$ is stable for (NL_{LV}) (NOT asymptotically)

$\rightsquigarrow (\bar{x},\bar{y}) = \left(\frac{d}{c}, \frac{a}{b}\right)$ is a isolated minimum for W.

